

04-operations

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0.1 NOT Logical Operator

- Returns false if the result is true and returns true if the result is false.

```
[1]: not True
```

```
[1]: False
```

```
[2]: not False
```

```
[2]: True
```

```
[3]: not not True
```

```
[3]: True
```

```
[ ]: # If it is NOT "rainy" then I will wear my blue suede shoes.  
rainy = input("Is it rainy? (yes/no) ")  
if not rainy == "yes":  
    print("Putting on my blue suede shoes")
```

0.2 AND Logical Operator

- If both operands are true, the condition is true

```
[4]: True and True
```

```
[4]: True
```

```
[5]: True and False
```

```
[5]: False
```

```
[6]: False and False
```

```
[6]: False
```

```
[7]: False and True
```

[7]: False

```
[10]: alice = True
      bob = True

      # If Alice AND Bob are here, begin the party -- we must have them both
      if alice and bob:
          print("Lets get this party started!")
```

Lets get this party started!

0.3 OR Logical Operator

- If any of the operands are true, the condition is true

```
[11]: True or True
```

[11]: True

```
[12]: True or False
```

[12]: True

```
[13]: False or False
```

[13]: False

```
[14]: False or True
```

[14]: True

```
[15]: alice = False
      bob = True

      # If Alice OR Bob are here, begin the party -- we can go with either
      if alice or bob:
          print("Lets get this party started!")
```

Lets get this party started!

0.4 XOR Bitwise Operator

```
[16]: True ^ True
```

[16]: False

```
[17]: True ^ False
```

[17]: True

```
[18]: False ^ True
```

[18]: True

```
[19]: False ^ False
```

[19]: False

```
[24]: alice = True
      bob = False

      # If Alice shows up, or if Bob shows up, begin the party,
      # but don't begin if nobody is there or if both show up
      # -- we don't want both those crazy characters at our party together
      if alice ^ bob:
          print("Lets get this party started!")
```

Lets get this party started!

```
[ ]:
```